Lab 03 - Prompt and Specification Pack

Prompt Experiment Matrix for Controlled Image Generation Generative AI for Image and Video Creation (TGS-2020505925) | Version v11.0 | 6 September 2026

This is the reusable prompt pack for the experiment. Cells A to D are written so that several factor groups may change per step; this is a progressive-specification comparison. Model IDs and parameters below were verified against the official Gemini API documentation on 6 September 2026; the imagen-4.0-* IDs that appear in older tutorials are deprecated and are not used here.

Currency note. Model identifiers and API parameters quoted anywhere in this pack were verified against the official vendor documentation on 6 September 2026. Model IDs change; re-verify before relying on one. Identifiers known to be deprecated are marked as such rather than quietly removed, so you can recognise them in older material.

Cell A - subject only (under-specified baseline)

```
A person wearing the Harbour Line jacket.
```

Everything else - setting, time of day, framing, lens, lighting and register - is left to the sampler. This is the condition the marketing team is complaining about.

Cell B - subject named + context named

```
A person wearing the Harbour Line jacket, standing on a harbour boardwalk at first light, with
mooring bollards and rigging behind them.
```

Cell C - subject LOCKED + context named + style named

```
Subject: the Harbour Line jacket, slate-blue shell with a single chest seam and a stand collar,
 worn open over a plain cream tee by an adult model facing three-quarters to camera.
Context: harbour boardwalk at first light, mooring bollards and rigging behind.
Style: documentary product photography, restrained colour, no lens flare.
```

Cell D - all four factors LOCKED, as structured JSON

```
{
  "prompt_version": "v3",
  "brief_reference": "NQ-HARBOUR-LINE-AW26",
  "subject": "the Harbour Line jacket: slate-blue shell, single chest seam, stand collar, worn
open over a plain cream tee by an adult model, three-quarters to camera, hands at sides",
  "context": "harbour boardwalk at first light; mooring bollards and rigging in the mid-ground;
 wet timber underfoot",
  "style": "documentary product photography, restrained colour palette, natural contrast",
  "technical": {"lens": "35mm", "depth_of_field": "shallow",
                "lighting": "low-angle warm key from camera left, soft fill",
                "aspect_ratio": "4:5", "image_size": "2K"},
  "negative_constraints": ["no text or logos rendered in the image",
                            "no lens flare", "no additional people",
                            "no colour cast on the jacket shell"]
}
```

One key per factor is the point: to test a tighter framing you change lens alone and the other three input fields stay unchanged; the generated appearance is not guaranteed unchanged.

Optional API call - current documented pattern (verified 6 Sep 2026)

Only run this if you have your own free-tier key. It is NOT required to complete the lab.

Before using this optional route, install a current google-genai package online, set GEMINI_API_KEY privately in your environment, and prepare one JSON prompt per bundle before scoring. Access/quotas are account-dependent. Run once for each of four variants per bundle, changing prompt path and output name. No seed control is assumed.

```
from google import genai
from pathlib import Path
import base64, json

Path('out/gen').mkdir(parents=True, exist_ok=True)
client = genai.Client()
spec = json.load(open('out/prompt-D.json'))

interaction = client.interactions.create(
    model="gemini-3.1-flash-image",   # current; imagen-4.0-* is deprecated
    input=json.dumps(spec),
    response_format={"type": "image", "aspect_ratio": "4:5", "image_size": "2K", "mime_type":
"image/png"},
)

if not interaction.output_image: raise RuntimeError('No image returned; record refusal and use
offline route')
open('out/gen/cellD_01.png', 'wb').write(
    base64.b64decode(interaction.output_image.data))
```

Documented aspect ratios include 1:1, 3:2, 2:3, 3:4, 4:3, 4:5, 5:4, 9:16, 16:9 and 21:9; output sizes are 1K/2K/4K depending on the model. All generated images carry a SynthID watermark - record that in your provenance note.

Governance note to attach to any generated asset

```
asset_id:         <file name>
generated_by:     <model id> | SIMULATED CLASSROOM ASSET
prompt_version:   v3  (out/prompt-v3.json, sha256 <hash>)
brief_reference:  NQ-HARBOUR-LINE-AW26
watermark:        provider documents SynthID; not independently detected | n/a (simulated)
reviewed_by:      <name>   review_date: <date>
```
